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Appendix

A General Experimental Setting

Target Model. Unless otherwise specified, we start from a generic instruction-tuned LLM (e.g., Llama-3.1-8B-Instruct [9]). The model is standard pre-trained and supervised fine-tuned, but is not specifically optimized for long chain-of-thought reasoning.

Data Source. We randomly sample 20K–35K high-quality Long CoT queries from OpenThoughts-3 [24] as the primary training corpus. The dataset provides multi-step reasoning traces, averaging over 20 steps per example, and covers diverse mathematical and logical problem types.

Evaluation Benchmarks. We evaluate on 6 challenging mathematical reasoning benchmarks that require multi-step logic:

- **GSM8K [25]:** Grade-school math problems requiring multi-step calculations.
- **MATH-500 [26]:** High-school and early college-level problems across major math topics.
- **AMC 2023 [27]:** High-school competition problems in algebra, geometry, and combinatorics.
- **AIME 2024 [28]:** Invitational-level problems with integer final answers.
- **AIME 2025 [29]:** Invitational-level problems with integer final answers.
- **OlymBench [30]:** A comprehensive Olympiad-level benchmark.

Unless stated otherwise, (Overall) Accuracy is the mean accuracy across all benchmarks above.

Inference and Metrics. For evaluation, we use the same sampling temperature as in reinforcement learning, $= 0.6$, to reduce the SFT–RL mismatch. We report Avg@1 accuracy for GSM8K, MATH-500, and OlymBench. For smaller test sets (AMC 2023 and AIME 2024/2025), we report Avg@16 accuracy. We instruct the model to output the final answer in \boxed{·} format to simplify extraction. We then parse boxed answers and match them to reference solutions using standard answer-matching tools (e.g., `math-verify`). Except for specifically specified variables, all model training settings are the same.

B Detailed Experimental Settings for Preliminary Study

In this appendix, we detail the experimental setup for the preliminary study on cold-start Long CoT. We compare 3 data construction pipelines. For all pipelines, we fine-tune the model for 1 epoch with a learning rate of $2e-5$ and a global batch size of 128, using a max sequence length of $16K$ or $32K$ tokens (we select the better-performing setting between these two lengths). We use Qwen2.5 [31] and Llama3.1 [9] as backbones to cold-start Long CoT across multiple model sizes.

Setting 1: Distillation from Strong Reasoning LLMs. To construct a high-quality synthetic dataset, we use a reasoning LLM as the teacher. Specifically, we distill Long CoT reasoning traces from DeepSeek-R1-671B-0528 [8], QwQ-32B [32], OpenAI-OSS-120B [33].

Setting 2: Distillation from Weak Instruction LLMs (ICL-Distill). To simulate surface imitation via in-context learning in instruction-tuned LLMs, we use a standard instruction model (e.g., Qwen2.5-32B-Instruct) that is not optimized for deep reasoning. We randomly select 1-shot exemplar (an R1-generated Long CoT trace) in the prompt, and generate solutions for the same problems as in Setting 1.

Setting 3: Fine-tuning on Human-Annotated Traces. We utilize 50 R1-generated reasoning traces and 50 human-written step-by-step solutions for supervised fine-tuning to compare against human-annotated reasoning traces. The data are collected from Du et al. [7].

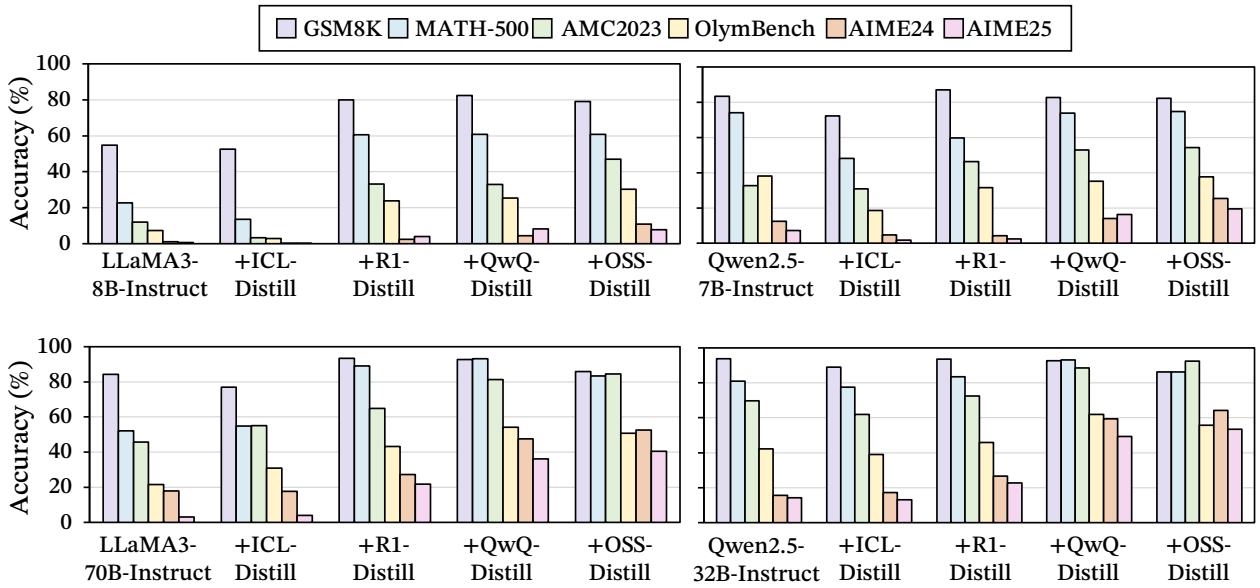

Figure 15 The full failure result of ICL from weak instruction-following LLMs and Human-annotated reasoning traces to acquire Long CoT structures, compared to successful distillation from strong reasoning LLMs.

C Reasoning Bond in Long CoT

In this section, we provide detailed experimental settings for analyzing reasoning bonds in Long CoT.

C.1 Mathematical Definition & Analysis of Reasoning Bonds

C.1.1 Reasoning Behavior Definition.

Given an input query x, a model generates an output text y containing intermediate reasoning followed by a final answer. We segment y into T step-level units ("steps") using standard delimiters (e.g., line breaks or bullet markers), following prior work [19, 34]. This yields a Long-CoT trajectory

$$\tau \coloneqq (u_1, \ldots, u_T), \tag{3}$$

where each u_t is a textual step.

To analyze the geometry of τ, we map each step u_t to a vector $h_t \in \mathbb{R}^d$. Concretely, we encode u_t with a fixed reference encoder and obtain token hidden states from a fixed layer (e.g., the penultimate layer). We then average token hidden states to form a step embedding h_t, producing an embedding sequence $(h_1, \ldots, h_T)$.

Behavior space and labeled transitions. We model Long-CoT as a sequence of step-to-step transitions. For each $t \in \{1, \ldots, T-1\}$, define the directed transition (edge)

$$e_t \coloneqq (u_t \to u_{t+1}). \tag{4}$$

Let the behavior label set be

$$\mathcal{B} \coloneqq \{\mathcal{N}, \mathcal{D}, \mathcal{R}, \mathcal{E}\}, \tag{5}$$

where $\mathcal{N}$ denotes Normal Operation, $\mathcal{D}$ Deep Reasoning, $\mathcal{R}$ Self-Reflection, and $\mathcal{E}$ Self-Exploration. An automated classifier (Appendix C.1) assigns a label $b_t \in \mathcal{B}$ to each transition e_t, yielding a labeled transition sequence $(e_t, b_t)_{t=1}^{T-1}$.

We formalize a behavior-graph framework for Long CoT, in which molecular-inspired 'bonds' correspond to behavior-labeled edges whose distributions can be estimated, compared, and transferred across models.